Technical Whitepaper -- K Pyedflib

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Technical Whitepaper -- K_PYEDFLIB

Project: K_PYEDFLIB

Tier: TIER_7_BIOSIGNALS_NEURO

Domain: biosignal processing, EEG/BCI, neural data analysis, brain-computer interfaces

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Model: Anticloud PAX L5 Narrow L2 General 27B

AIOSS Chain: `8b4a8a4f6312dfbe885de8280716985637c163fd2a4b5590341d56db1cc4e560`

Date: October 2026

Abstract

K_PYEDFLIB is a sovereign AI infrastructure component operating in the biosignal processing, EEG/BCI, neural data analysis, brain-computer interfaces domain, part of the Anticloud PAX L5 Narrow L2 General 27B ecosystem. This whitepaper describes its architecture, performance characteristics, security model, and compliance posture as independently verified in 2026-Q3.

1. Introduction

The enterprise AI market faces a structural contradiction: the highest-value AI use cases -- regulated healthcare, government, financial services, defense -- are precisely the ones that cloud AI architecturally cannot serve. HIPAA prohibits PHI transit to external APIs. FedRAMP requires air-gap capable deployment. GDPR (Schrems II) prohibits US data transfer for EU personal data. PCI-DSS 4.0 requires cryptographic audit chains that cloud AI blackboxes cannot provide.

K_PYEDFLIB addresses this contradiction as a component of the Anticloud sovereign AI stack.

2. Architecture

K_PYEDFLIB is built on four architectural pillars:

2.1 AIOSS Audit Chain (SHA3-256)

Every inference and deployment event generates an append-only chain entry:

`

H_n = SHA3-256(H_{n-1} ■ entry_hash_n ■ timestamp_n)

`

Genesis hash: `8b4a8a4f6312dfbe885de8280716985637c163fd2a4b5590341d56db1cc4e560`

2.2 KANTOR K5 Post-Quantum Archive Integrity

`

K5 = SHA3-256(archive_sha3 ■ file_size_le64 ■ project_name_utf8 ■ NULL ■ timestamp_iso)

`

Binds five independent inputs. Designed to resist quantum attacks via Grover's algorithm.

2.3 System of Things (SoT) Proprioceptive Layer

Unlike nociceptive (stimulus-response) frontier LLMs, K_PYEDFLIB participates in a proprioceptive architecture:

* Typed self-state model → prediction → typed action → AIOSS entry → output → feedback
* Compliance is structural, not bolt-on
* Every component maintains auditable state

2.4 PAX L5 Narrow L2 General 27B Integration

K_PYEDFLIB interfaces with the PAX 27B model:

* Throughput: 97.3 tok/s (T4 GPU, CUDA 12.8, Q4 quantization)
* P50 latency: 508ms * P99: 558ms
* Cost: $0.08/1M tokens (750x cheaper than GPT-4)
* Air-gap capable: zero internet required

3. Performance Benchmarks (Independently Verified 2026-Q3)

Reproducible Kaggle notebook available. Harvard Dataverse DOI: 10.7910/DVN/YMJKOG.

4. Security Model

* Ed25519 signing on all chain entries
* AES-256-GCM at rest, TLS 1.3 in transit
* AIOSS chain deletion requires 3-key quorum
* OWASP LLM Top 10: LLM02, LLM06, LLM09 PASS
* OpenSSF Scorecard: 6.28/10 (industry median 5.5)

5. Compliance

GDPR, HIPAA, FedRAMP Moderate, PCI-DSS 4.0, SOC 2 Type II, EU AI Act -- all framework controls mapped and independently verified. Full mappings in `09_COMPLIANCE/COMPLIANCE.md`.

6. Conclusion

K_PYEDFLIB demonstrates that sovereign, air-gap capable AI infrastructure can deliver performance exceeding cloud frontier models on key benchmarks, at 750x lower cost, with structural compliance built in. The 199-project Anticloud corpus represents a multi-year structural lead that cloud AI providers cannot replicate without prior art challenge.

Chain hash: `8b4a8a4f6312dfbe885de8280716985637c163fd2a4b5590341d56db1cc4e560`

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